

PUBLICLY AVAILABLE SPECIFICATION

PRE-STANDARD



**Smart manufacturing service platform – Service-oriented integration
requirements of the manufacturing resource/capability**



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INTERNATIONAL ELECTROTECHNICAL COMMISSION

SMART MANUFACTURING SERVICE PLATFORM – SERVICE-ORIENTED INTEGRATION REQUIREMENTS OF THE MANUFACTURING RESOURCE/CAPABILITY

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The text of this PAS is based on the following document:

This PAS was approved for publication by the P-members of the committee concerned as indicated in the following document

Draft PAS	Report on voting
65E/578/DPAS	65E/585/RVDPAS

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SMART MANUFACTURING SERVICE PLATFORM – SERVICE-ORIENTED INTEGRATION REQUIREMENTS OF THE MANUFACTURING RESOURCE/CAPABILITY

1 Scope

This PAS provides the requirements of all relevant manufacturing resources integrated to the cloud manufacturing service platform, including integration of hard manufacturing resources, soft manufacturing resources and manufacturing capabilities.

This document is used for the integration of the relevant resources to the smart manufacturing service platform.

2 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org/>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

2.1

soft manufacturing resource

manufacturing resources that are based on software, data, models, knowledge

2.2

hard manufacturing resource

manufacturing equipment, computing equipment, materials, etc. used in a product life cycle

2.3

manufacturing capability

ability to complete various activities during the product life cycle, featured by the organic combination of three main elements, human labour, management and technology

3 General requirements

The following requirements should be met:

- a) Resource/capability integration to the smart manufacturing platform is generally divided into hard manufacturing resource integration, soft manufacturing resource integration and manufacturing capability integration.
- b) Status data of soft/hard manufacturing resource and capability should be extracted, and integrated to smart manufacturing platform through a wired or wireless network.
- c) Physical manufacturing resource/capability should be transformed to logical ones. Logical manufacturing resource/capability should be defined by unified description, which forms description document of virtual manufacturing resource/capability.
- d) Virtual manufacturing resource/capability should be integrated to the smart manufacturing platform as a cloud service through a unified description of services and packaging method.

4 Integration requirements of hard manufacturing resources

4.1 Integrated content

According to the business requirements in a manufacturing life-cycle, integrated hard manufacturing resources typically include the following:

- a) complete set of equipment, such as large-scale simulation test system and digital production line, etc.;
- b) production equipment, including mechanical equipment, electrical equipment, environmental test equipment, etc.

4.2 Application requirements after integration

Integrated equipment/device should be applied in the following ways:

- a) equipment/device status monitoring and resource utilization analysing;
- b) equipment/device management;
- c) equipment/device online renting and calling;
- d) resource collaborative planning.

Integrated equipment/device should provide following information:

- 1) equipment/device general information;
- 2) equipment/device capability information;
- 3) equipment/device time information;
- 4) equipment/device status information;
- 5) equipment/device task information.

4.3 Integration mode

4.3.1 Complete set of equipment

4.3.1.1 Complete set of simulation test equipment

For the set of simulation test equipment which is well digitalized and has a complete set of control systems, a local server with dual DNC can be added to connect and convert real time network (reflective memory network) to research network, so that device is integrated and instrumentation is realized, as shown in Figure 1.

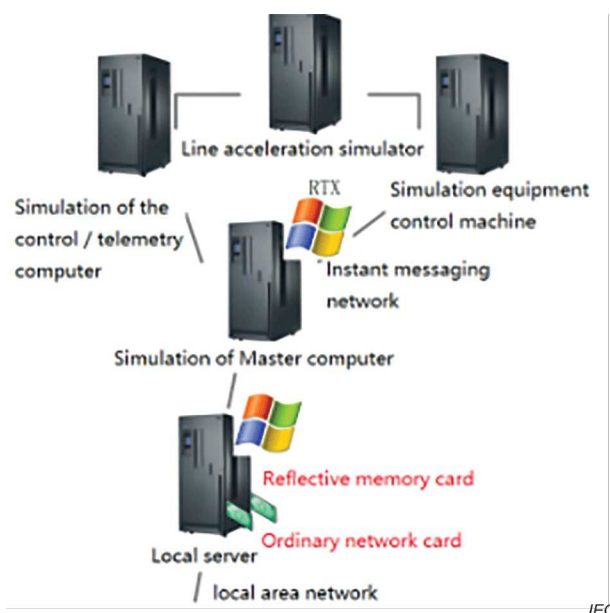


Figure 1 – Integration method for complete set of simulation equipment

4.3.1.2 Digital production line

For digital product line, there are generally two ways of integration:

- integrating distributed numerical control (DNC) centers, manufacturing execution systems (MES) and enterprise resource planning (ERP) systems to realize high-end integration and control of digital production line;
- integrating digital production line through digital manufacturing center database, which is generated by fully or semi-automatically collecting data of working status for equipment, which is realized by adding bar code labels to tools and materials and giving unified digital labels to all kinds of production resources.

4.3.2 Production equipment

Production equipment integration has the following two situations based on whether it has networking capabilities or not:

- a) Install smart capture terminals and sensors or I/O signal capture devices on the production device which does not have the networking capacity, and then equipment should be integrated.
- b) There are generally two types of integration for production equipment that has a networking capacity:
 - 1) integrated through deploying industrial data gateways which has a platform integration functionality;
 - 2) integrated through digital numerical control centers (DNC), manufacturing execution systems (MES) and enterprise resource planning systems (ERP).

5 Integration requirements of soft manufacturing resources

5.1 Integrated content

Integration of soft manufacturing resources should be able to implement the resource management and on-demand usage. Soft manufacturing resources which are integrated to the smart manufacturing cloud platform typically include the following:

- a) enterprise information systems: including ERP, CRM, PLM, OA, etc.;

- b) tooling software: including CAD, CAE, CAM, CAPP, EDA, etc.;
- c) application integration platforms: including collaborative simulation platform, multidisciplinary optimization platform, etc.;
- d) knowledge base: including public tooling libraries, parts libraries, fault diagnosis rule base, etc.

5.2 Requirements of application mode after integrated

5.2.1 Batch mode

Batch mode requirements of integrated soft manufacturing resources include the following:

Multi-agent users should be able to submit their compute-intensive analysis tasks (such as complex aerospace products pneumatic and thermal field analysis) or production tasks respectively in batch.

- a) The cloud-based intelligent manufacturing platform should provide suitable problem-solving resources/capabilities to promote efficiency significantly for multi-agent users.
- b) The cloud-based intelligent manufacturing platform should be able to prioritize tasks in a queue, so the user can focus on other tasks first.
- c) It should show users the status, process and result of a task.

5.2.2 Virtual interactive mode

Virtual interactive mode requirements of integrated soft manufacturing resources include the following:

- a) Multi-agent user should hand in their customized requirements on manufacturing resources/capabilities or environment of resources to cloud-based smart manufacturing platform.
- b) Multi-agent users can easily get and operate cloud environment to finish the manufacturing tasks such as structural design, process design and even the semi-physical simulation test, etc.
- c) The cloud-based smart manufacturing platform should be able to help create virtualized resource/ capability environment and transmit operating interface to user desktop.
- d) The user doesn't have to understand where the environment of resources/capabilities locates and the ways to maintain the environment, so that he can focus on its expertized business area in the virtual interactive mode.

5.2.3 Collaborative interoperability mode

Collaborative interoperability mode requirements of integrated soft manufacturing resources include the following:

- a) Multi-agent users should be able to collaborate on manufacturing tasks based on different manufacturing resources/capabilities (such as multidisciplinary design optimization for virtual prototyping, semi-physical simulation test, etc.)
- b) The platform should be able to create a collaboration environment of same time/space automatically. In a wide-area distributed environment (or part of the high-performance computing cluster environment), interoperability can be automated, or business processes can be transferred, manufacturing resources/ability can be dynamically looked up, called and synchronized to finish certain tasks.
- c) Users should be able to check the task status as well as to track the intermediate results of collaborative manufacturing job in the cloud through 3D visualization, etc. based on noted collaborative processes.

5.3 Integration method

Integration requirements of soft manufacturing resources include the following:

- a) Virtualized and service-oriented reform of the integrated soft manufacturing resources.
- b) Use virtual application publishing or service-oriented (SOA, SaaS) packaging to integrate soft manufacturing resources.
- c) Turn on premise software module into the cloud version before virtualized and service-oriented reform.
- d) To EXE software or module programmed by VC++ in Windows system, automation technology can be used to expose the interface similar to DLL.
- e) Integration can be realized by Jacob which uses COM called by Java, under the condition that the soft resources are packaged to Web Services through J2EE.
- f) Reform the browser/server (B/S) structured software and module to user interface layer, business logic layer and data access layer to satisfy virtualized and service-oriented requirements from multiple tenants.

6 Integration requirements of manufacturing capabilities

6.1 Integration content

Manufacturing capabilities which are integrated to the smart manufacturing cloud platform typically include the following:

- a) Human labour and relative business logic: such as approval process of warehouse manager, etc.
- b) Enterprise internal information system: including integration between different ERP systems as well as integration between cloud manufacturing system and enterprise internal information systems.
- c) Classification of manufacturing capabilities service (refer to Clause A.2).

Generally, there are two types of classification of manufacturing capabilities.

- a) In accordance with the manufacturing life cycle activities, including research and development, supply, production, marketing, services, as shown in Figure 2.

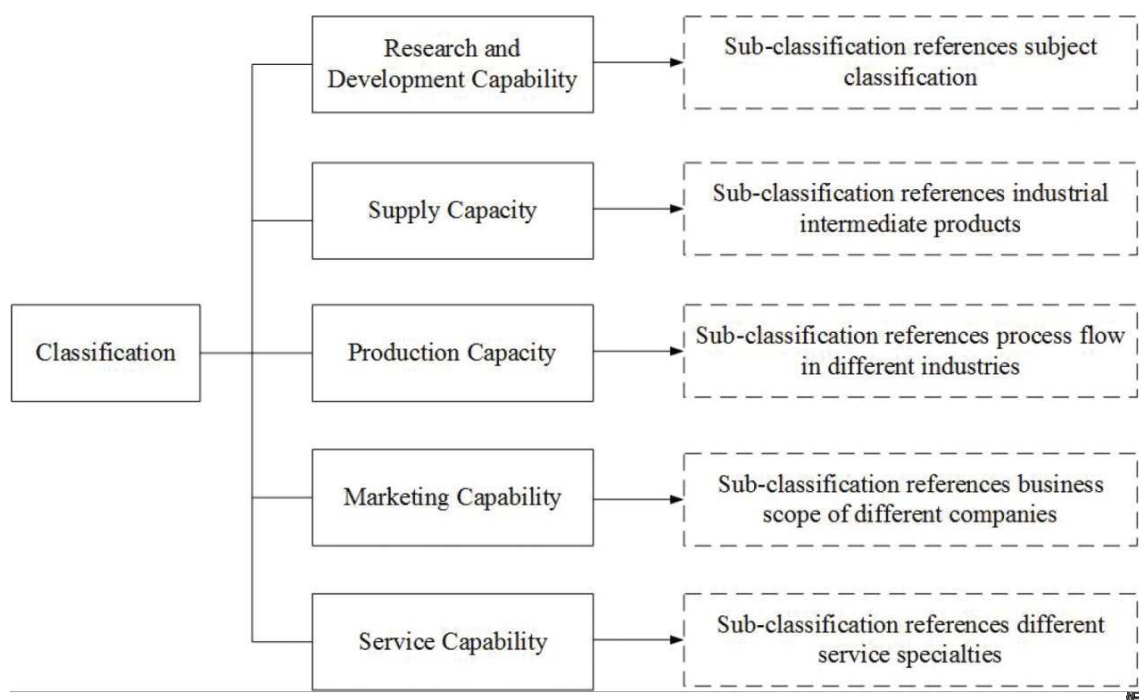


Figure 2 – Capability classification based on product life cycle

- b) In accordance with the types of equipment manufacturing, including general equipment manufacturing, special equipment manufacturing, transport equipment manufacturing, electrical machinery and equipment manufacturing, communications equipment, computers and other electronic equipment manufacturing, instruments and other mechanical manufacturing capabilities, as shown in Figure 3.

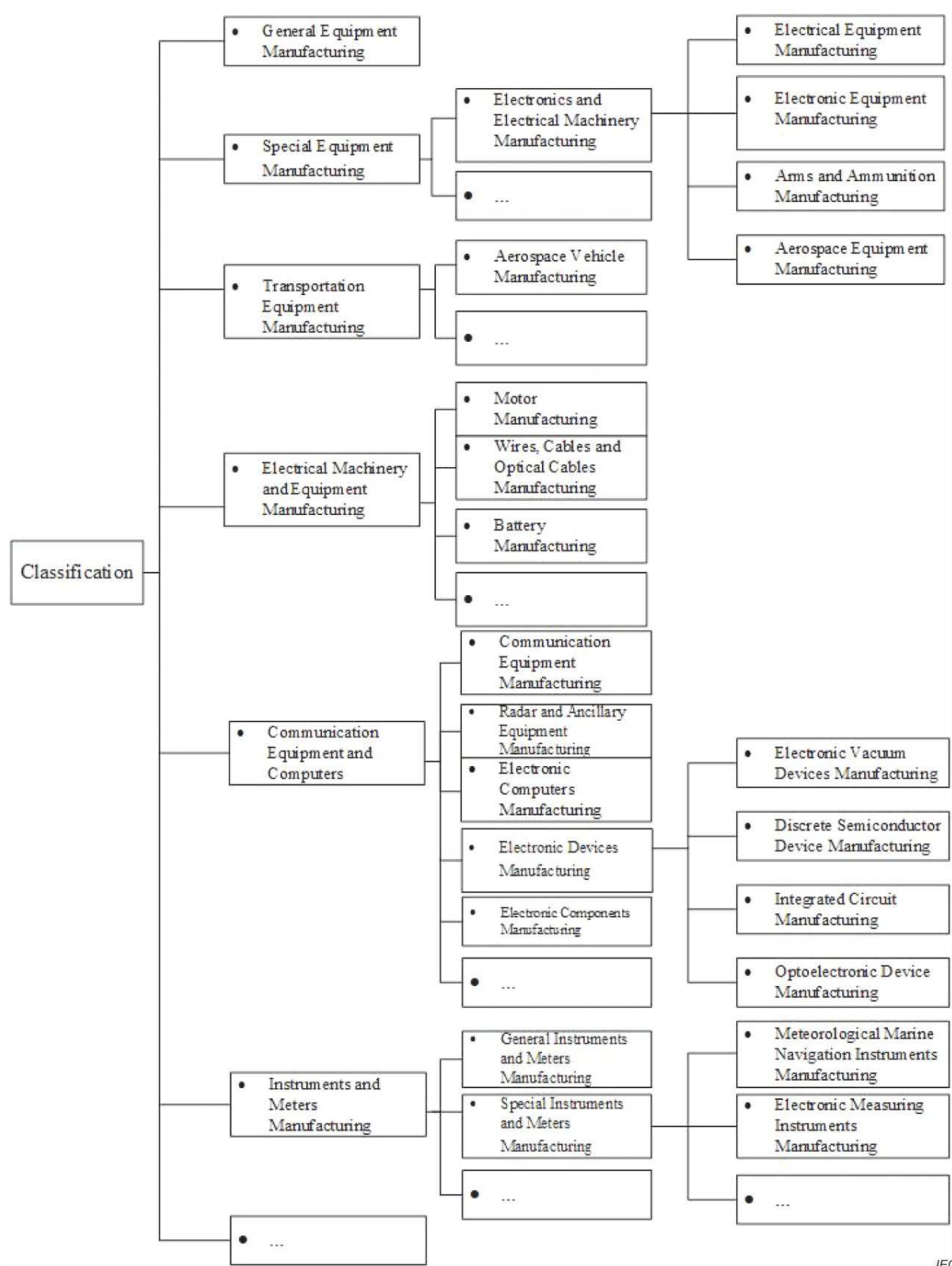


Figure 3 – Capability classification based on types of equipment manufacturing

6.2 Requirements of application mode after integrated

Multi-agent users utilize relevant manufacturing capabilities, such as knowledge and professional sites based on their own requirements, to do creative work in regards to design, simulation, production, test, integrating on-demand manufacturing capabilities.

Smart manufacturing platform should provide knowledge/capability management functionality. It should be able to coordinate teams and devices in different subjects and systems and implement registration, publication, dynamic tracking and on-demand integration of manufacturing capabilities.

6.3 Integration mode

Requirements of integration mode of manufacturing capabilities are listed as follows:

- a) Registration: the provider should have a globally unique ID number, customized implementation method, monitoring and distribution optimization functionality;
- b) Key features: the provider should have a scale of enterprise assets, personnel as well as technology maturity and qualification;
- c) Usage method: the provider is encouraged to provide types of services, such as online transfer, cloud scheduling and electronic collaboration. Service can be provided through Software as a Service (SaaS) to multi-agent users.

7 Classification of cloud manufacturing service

Based on the above hard and soft manufacturing resources and integration requirements of manufacturing capabilities, cloud manufacturing services are classified (see Clause A.3).

Based on the hard manufacturing resources, soft manufacturing resources and manufacturing capabilities, cloud manufacturing service platform application layers are classified (see Clause A.3).

Annex A (informative)

Resource service classification

A.1 Resource service classification (see Figure A.1)

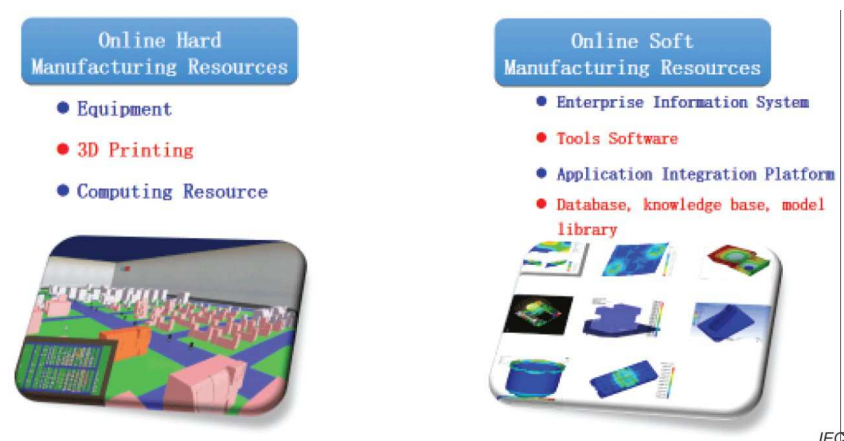


Figure A.1 – Resource service classification

A.2 Classification of manufacturing capabilities service (see Figure A.2)



Figure A.2 – Classification of manufacturing capabilities service

A.3 Classification of cloud manufacturing service (see Figure A.3)

Classification of service-The whole industry chain cloud manufacturing services

- Cloud Customization and R&D
 - User Deep Intervention Model
- Cloud Collaboration R&D Service
 - Crowdsourcing Model
 - Simulation Model of Online Collaboration Design
 - Integration Model of Online Design and Manufacture
 - Collaborative Simulation Analysis
- Cloud Collaboration Procurement Service
 - Bidding
 - Inquiry and Quotation
 - Director Order
- Cloud Collaboration Production Service
 - Subcontract
 - Collaboration Scheduling
 - 3D Printing
- Cloud Collaboration Marketing Service
 - Agency Model
 - Distribution Model
 - Associated Model
- Cloud Collaboration After-sales Service and Value-added Service
 - Door-to-door Service
 - On-site Technical Service
 - Value-added Service



Service Classification- Supporting Services



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Figure A.3 – Classification of cloud manufacturing service

Annex B (informative)

Introduction of cloud manufacturing service platform

B.1 Background

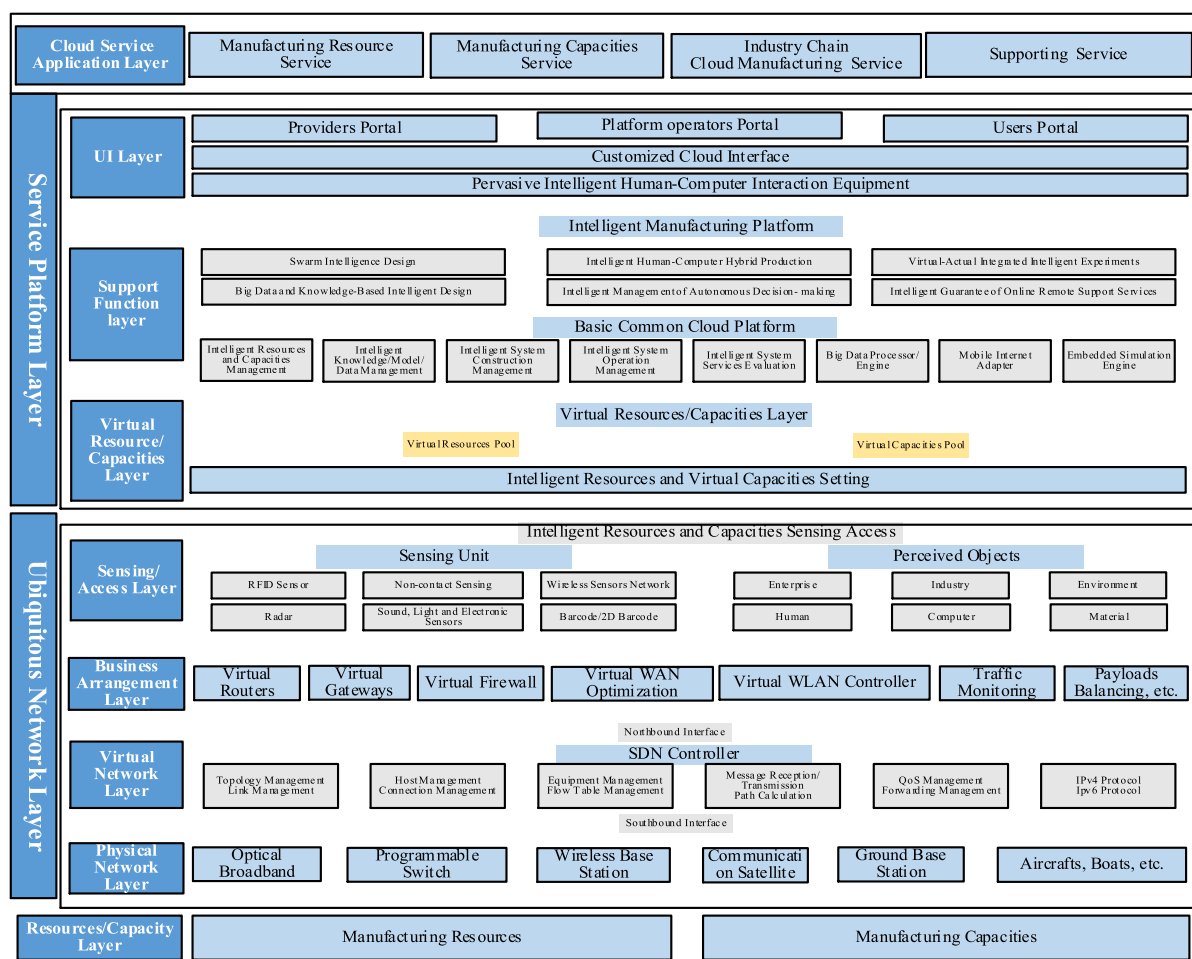
Cloud manufacturing is a new manufacturing model by which new information and communication technology, intelligent science and technology, large manufacturing (design/production/management/testing and integration) technology, system engineering technology and related product technology are integrated in the whole system and lifecycle of product development, so as to realize autonomous sensing, interconnection, collaboration, learning, analysis, cognition, decision-making, control and execution of human, machine, material, environment and information in the life cycle of manufacturing, to enable the integration and optimization of people/organization, operation management, equipment and technology (three elements) of manufacturing enterprise (or group) as well as information flow, logistics flow, capital flow, knowledge flow and service flow (five flows), to facilitate the products production and serving the users with a high efficiency, a high quality, and in a cost-effective and environmental-friendly manner, and thus to improve the market competitiveness of manufacturing enterprises (or group).

Cloud manufacturing can constitute the service cloud of manufacturing resources and capabilities with a unified management so that users will be able to obtain manufacturing resources and capabilities on demand through the cloud manufacturing service platform. Internet-based, service-oriented, collaborative, customizable, flexible and socialized intelligent manufacturing system is used to facilitate products production and provide services to users.

Cloud manufacturing service platform is the carrier of the cloud manufacturing, and also an industrial cloud platform. It has the following characteristics:

- **Digital:** The attributes of manufacturing resources and capabilities, as well as static/dynamic behaviour, are transformed into numbers, data, and model for unified analysis, planning, and reorganization.
- **Instrumented:** The integration of information technology and manufacturing technology, such as Internet of cyber-physical system (CPS), comprehensive integration and perception of soft and hard manufacturing resources and capabilities can be realized.
- **Virtual:** Provide logical and abstract representation and management of manufacturing resources and capabilities, without physical constraints.
- **Service-oriented:** Virtual intelligent manufacturing resources and capabilities are encapsulated and assembled through service computing technology to form manufacturing services.
- **Collaborative:** Providing comprehensive support for swift dynamic collaborative virtual enterprise management to enable seamless integration of multi subject (tenant) on-demand dynamic construction of virtual enterprise organization and virtual enterprise business operation.
- **Customizable:** Enabling on-demand acquisition of intelligent manufacturing resources and capabilities at any time and place, to support personalized and customized manufacturing.
- **Flexible:** Support the flexible and interconnected manufacturing activities of people, machines and things.
- **Intelligent:** Integrates human, intelligent science and technology, big data, CPS and simulation technology and manufacturing technology into the entire manufacturing system and lifecycle to self-sense, learn, analyse, recognize, decide, control and execute the person, machine, material, environment, information in the system.

B.2 Architecture of cloud manufacturing service platform (see Figure B.1)



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Figure B.1 – Architecture of cloud manufacturing service platform

Function requirements of each layer of architecture of the cloud manufacturing service platform:

- 1) Resources/capacities layer. Including manufacturing resources and manufacturing capacities, such as hard manufacturing resources like machine tools/robots/machining centers/computing equipment /simulation test equipment, materials and energies; soft manufacturing resources like models, (big) data, software, information and knowledge in the manufacturing process; manufacturing capacities of demonstration, design, production, simulation, experiment, management, sales, operation, maintenance and integration in the manufacturing process as well as new digital, networked and intelligent manufacturing interconnection products.
- 2) Ubiquitous network layer. Consisting of physical network layer, virtual network layer, business arrangement layer and intelligence sensing/access layer.
 - Physical network layer. Mainly including optical broadband, programmable switch, wireless base stations, communication satellites, ground base stations, aircraft and boats, etc.
 - Virtual network layer. Achieving open network through southbound and northbound interfaces for topology management, host management, equipment management, message reception/transmission, QoS management and IPv4/Ipv6 protocol management.
 - Business arrangement layer. Providing the network functions in the form of software through software and hardware decoupling and function abstracting to achieve a rapid development and deployment of new businesses and providing virtual router, virtual

- firewall, virtual WAN optimization and control, traffic monitoring and payloads balancing, etc.
- Sensing/access layer. Sensing the objects such as enterprise, industry, human, machine and material through the intelligent sensing units of RFID sensor, wireless sensor network, sound, light and electronic sensors/equipment, bar code/2D bar code and radar, and transferring data and instructions through network.
- 3) Service platform layer. Consisting of virtual intelligent resources/capacities layer, core intelligent support function layer and intelligent UI layer.
- Virtual resources/capacities layer. Intelligent describing and virtual setting manufacturing resources/capacities to map the physical resources/capacities into logical intelligent resources/capacities to form virtual intelligent resources/capacities pool;
 - Core support function layer. Consisting of basic common cloud platform and intelligent manufacturing platform respectively providing the base middleware functions such as intelligent system construction management, intelligent system operation management, intelligent system service evaluation as well as intelligent manufacturing functions such as swarm intelligence design, big data and knowledge-based intelligence design, intelligent human-computer hybrid production, virtual-actual combination intelligent experiment, intelligent management of autonomous decision-making and intelligent guarantee of online service remote support;
 - UI layer. Pervasively supporting intelligent terminal interaction equipment for service providers, operators and users to achieve customized user environment.
- 4) Intelligent cloud service application layer. Provide manufacturing resource service, manufacturing capabilities service, the whole industry chain cloud manufacturing services and supporting services. The whole industry chain cloud manufacturing services includes cloud collaborative research and development, cloud collaborative procurement, cloud collaborative production, cloud collaborative marketing, cloud collaborative after-sales service and so on.

B.3 Cloud manufacturing services platform applications

B.3.1 Manufacturing industry cloud (see Figure B.2)

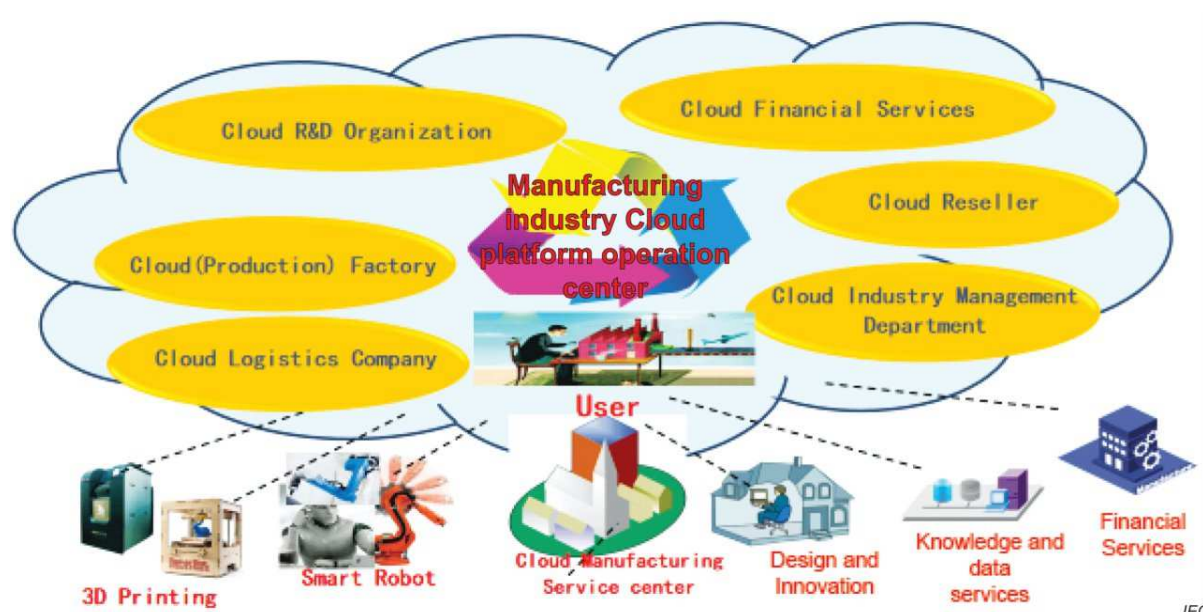


Figure B.2 – Manufacturing industry cloud

B.3.2 Manufacturing enterprise (factory) cloud (see Figure B.3)

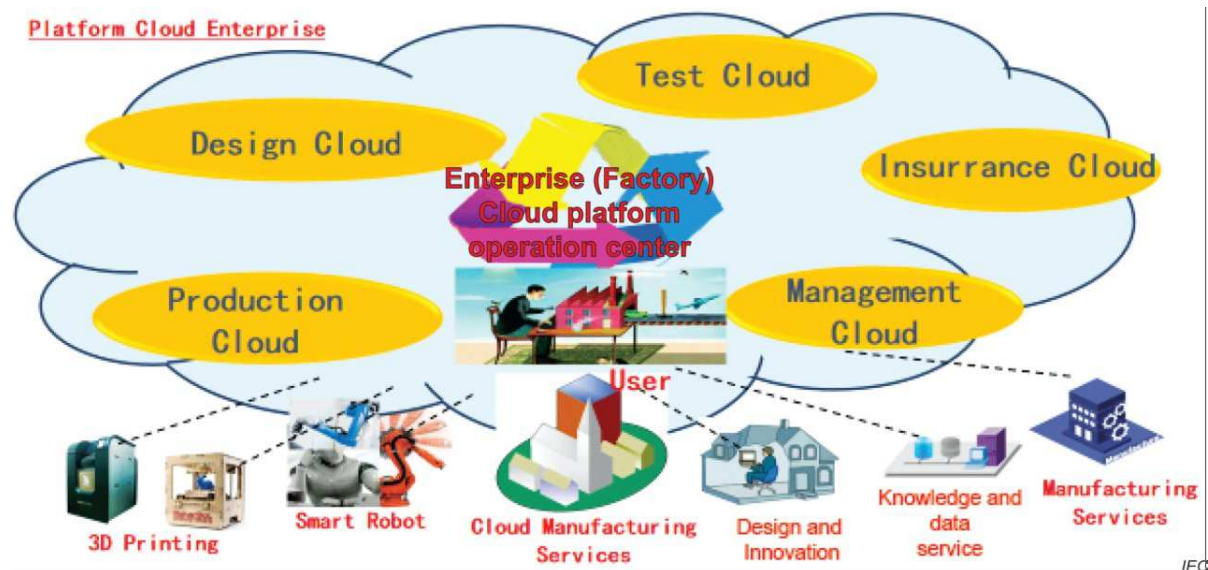


Figure B.3 – Manufacturing enterprise (factory) cloud

B.3.3 Manufacturing workshop cloud (production cloud) (see Figure B.4)

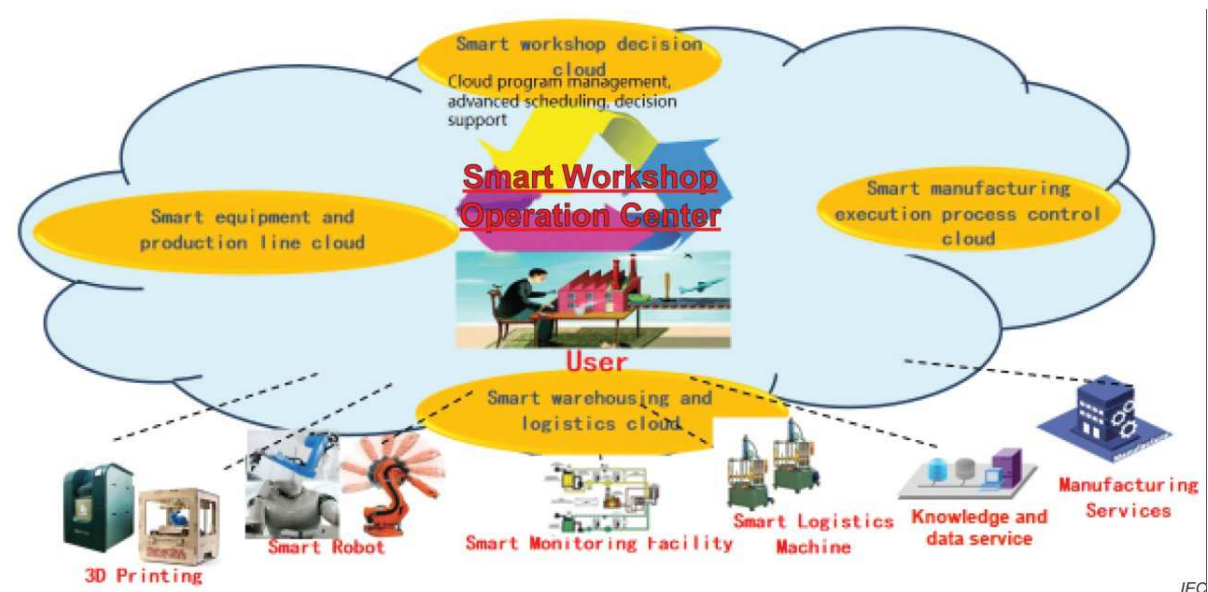


Figure B.4 – Manufacturing workshop cloud (production cloud)

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